



Decision Support

Portfolio optimization with robust stochastic dominance testing: A genetic algorithm approach

Jakub Neugebauer 

Prague University of Economics and Business, Department of Econometrics, W. Churchill Sq. 1938/4, Prague, 130 67, Czech Republic



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ABSTRACT

This paper introduces the s -RSD model, a robust portfolio optimization framework that generalizes stochastic dominance constraints of any given order s by allowing for outlier disturbances within the bounds of a Kolmogorov-Smirnov like statistical test critical value at a chosen significance level α . By generalizing strict dominance and allowing controlled, statistically insignificant violations, the model robustly balances return maximization with risk control. Genetic Algorithm is employed to solve the resulting nonlinear optimization problem, accommodating cardinality constraints and weight bounds, and achieving rapid convergence relative to exact solvers. Empirical analysis was performed on S&P 500 data under bearish, neutral, and bullish market conditions, demonstrating that, with appropriate tuning of the stochastic dominance order and significance level α , it is possible to reduce losses during downturns, enhance returns in stable markets, and realize outsized gains. Investor-specific tailoring of the risk-return trade-off is enabled by the adjustable parameters s and α . A flexible and powerful tool for modern portfolio management is provided by the proposed methodology.

1. Introduction

In uncertain times, it is important to develop a strategy to fight against negative financial phenomena such as inflation and thus protect our incomes from degradation. To do so, there are several factors that can guide our decision-making, the three most commonly used being return on investment, risk of investment, and liquidity. Even though many papers focus on all three factors, this paper aims to maximize return on investment while minimizing the risk of losing invested money, as liquidity is generally the least important factor in the investment triangle for investors.

Modern portfolio optimization theory has gone a long way since Markowitz (1952) laid its basis with his mean-variance model. This model's main disadvantage was its assumption of normally distributed returns, which is usually not met. There have been many proposals on how to avoid this problem, such as introducing skewness and kurtosis as other factors (e.g., (Abid et al., 2023; Zhao et al., 2015)), or using different risk measures such as semi-variance, CVaR, expected shortfall, and others. One can find a comparison between these and other risk measures in Ramos et al. (2023). Another example of handling the normality problem is assuming different distributions of returns, like the skew-Student distribution (see (Eling, 2014)). Among the discussed methodologies of portfolio optimization is fuzzy optimization (see, e.g., (Borovička, 2020; Li et al., 2010)). This method's benefit lies in incorporating vague information about asset returns and investor preferences.

Current approaches to gain precision in mean-risk models are machine learning techniques like long-short-term memory neural networks, random forests, XGBoost, and others (e.g., (Kwak et al., 2021; Ma et al., 2021; Wu et al., 2023)).

Despite the widespread use and accessibility of mean-risk models, they are not without challenges, particularly in the subjective selection of specific return or risk levels. This introduces an element of human error and ambiguity due to the absence of standardized rules for risk-measure and parameter selection. Consequently, this paper shifts its focus to the integration of stochastic dominance in portfolio optimization.

Rather than sticking to a specific risk measure, this study utilizes a constraint that forces the portfolio to stochastically dominate a chosen benchmark (typically an index) as a replacement for the Markowitz mean-risk model. This approach eliminates the need for subjective risk or return level selection and ensures that the portfolio is statistically superior to the benchmark based on the chosen order of stochastic dominance.

Current approaches to portfolio optimization using stochastic dominance constraints often involve statistical methods to formulate integer linear programming (ILP) or mixed-integer linear programming (MILP) models for a specific order of stochastic dominance (see, e.g., (Dentcheva & Ruszczyński, 2006) for the second order). Recently, robust approaches have been proposed to tackle the problem of selecting input data for the model and the resulting output bias (see, e.g., (Hosseini-Nodeh et al., 2022; Sehgal & Mehra, 2020)). These papers focus on the

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second order of stochastic dominance. However, this paper introduces a novel methodology: leveraging a statistical test of stochastic dominance at a significance level α to robustly determine whether the portfolio outperforms the chosen benchmark. This method not only offers flexibility in adjusting the desired order of stochastic dominance but also provides a mechanism for selecting a risk-return ratio through the significance level α , and it is generalized for any order s of stochastic dominance.

The goal of this paper is to develop a genetic algorithm (GA) that can solve a portfolio optimization model with a general-order stochastic dominance constraint, using the statistical test with a specific critical value. GA is suited for nonlinear problems, which aligns with the nature of portfolio optimization models. Its ability to optimize these problems makes it particularly effective when conventional linear programming and gradient-based methods struggle. GA has been widely applied and validated across various domains, including finance, engineering, and artificial intelligence (Kaucic et al., 2019). This widespread application provides a rich foundation of empirical evidence supporting its effectiveness in solving complex optimization problems.

The beginning of this paper introduces stochastic dominance, testing of stochastic dominance, and its use in portfolio optimization, with an overview of the recent literature in this field. In the core section of this paper, included in Section 3, the s -RSD model itself is proposed. Exact methods for solving it are discussed for any general order. Superconvex conditions are proposed and applied on order $s \geq 3$. The optimization technique used in this paper, genetic algorithm (GA), is described in Section 4. A weight-adjustment algorithm incorporated within this GA is proposed to solve the model with cardinality constraints, making the result of portfolio optimization applicable in real-life scenarios. This algorithm improves the convergence properties of the GA. The final part of this paper, Section 5, presents a simulation study designed to showcase the potential of the proposed method in comparison with the current literature.

2. Stochastic dominance

The concept of stochastic dominance is a valuable method in decision-making that compares two random variables based on their entire probability distributions. This provides more robust insight compared to traditional statistics such as mean, variance, VaR, and others, as it does not compare one selected value but compares all possible outcomes of random variables and decides which is “better.” Origins of the concept of stochastic dominance can be traced to Hadar and Russell (1969), Hanoch and Levy (1969), who formulated the basis for the widely used concept today.

Stochastic dominance is connected to the theory of the rational investor (Oehmsch, 1967), and there are multiple orders of stochastic dominance that have interpretations in the field of financial investments. First order stochastic dominance (FSD) is connected to the rule that the investor prefers higher returns on investment, a principle that remains fundamental in modern financial theory (Post & Levy, 2005). Second order stochastic dominance (SSD) reflects the risk aversion characteristic of investors, implying that a rational investor prefers lower risk for a given expected return on investment (Rothschild & Stiglitz, 1970). Third order stochastic dominance (TSD) captures the tendency of investors to prefer positively skewed distributions of return over negatively skewed ones, reflecting preferences for investments with the potential for high upside returns (Leshno & Levy, 2002). These various orders of stochastic dominance provide an essential framework for analyzing investment decisions in financial markets, considering the diverse risk preferences and behavioural patterns of investors. Stochastic dominance is defined as follows:

Let X and Y be random variables with cumulative distribution functions $F_X(x)$, respectively $F_Y(x)$. Let $D_X^1(x) \equiv F_X(x)$ and:

$$D_X^s(x) = \int_{-\infty}^x D_X^{s-1}(t) dt, \quad s \in \mathbb{Z}^+, \quad s \geq 2 \quad (1)$$

then we say that the random variable X stochastically dominates random variable Y at order s if $D_X^s(x) \leq D_Y^s(x)$ for all x . This is a case of weak stochastic dominance. In the case of strong stochastic dominance, we would need to add the condition that $D_X^s(x) < D_Y^s(x)$ for at least one x . In this paper, only the case of weak stochastic dominance will be considered, and the fact that random variable X stochastically dominates random variable Y at order s will be noted as $X \succeq_{(s)} Y$. It can be easily proved, that $X \succeq_{(s)} Y \implies X \succeq_{(s+1)} Y$. Higher orders of stochastic dominance are thus less constraining in portfolio optimization models.

2.1. Testing stochastic dominance

Testing stochastic dominance is the best way to robustly determine whether one random variable stochastically dominates another. The first tests were introduced by McFadden (1989) and Kaur et al. (1994). A robust stochastic dominance test for time series was proposed by Linton et al. (2005) and later improved by (Linton et al., 2010).

A broad range of approaches to stochastic dominance testing includes, for example, consistent Kolmogorov–Smirnov type tests developed by Barrett and Donald (2003), shape-restricted inference for Lorenz curves using duality theory introduced by Dentcheva and Penev (2010), and mean-risk-based dominance tests based on coherent risk measures proposed by Dentcheva et al. (2011), which have shown greater power than the Mann–Whitney test. Other refinements include weighted McFadden-type statistics for higher-order dominance (Horváth et al., 2006). Among more recent contributions is a recentering method applied to a continuum of inequality constraints, introduced by Donald and Hsu (2016). For readers interested in comparative performance, Tse and Zhang (2004) provide a Monte Carlo evaluation of several available test methods.

The following consistent stochastic dominance test, based on the one-way Kolmogorov–Smirnov metric, is taken from Linton et al. (2005), as it was built specifically for time series and fits the purpose of this paper. Another benefit of this test is that it is formulated generally for any given order of stochastic dominance s . The formulation of the test statistic, which will be used later as a portfolio optimization criterion, is given as follows.

Let the empirical distribution $\hat{F}_X(x)$ of variable X with sample $\hat{X} = \{X_1, X_2, \dots, X_I\}$ of size I be:

$$\hat{F}_X(x) = \frac{1}{I} \sum_{i=1}^I 1(X_i \leq x) \quad (2)$$

where $1(X_i \leq x)$ is a function that returns a binary variable that is equal to 1 if $X_i \leq x$ and 0 otherwise.

Let $\hat{\mathbf{R}}$ be the matrix of samples of portfolio (P) returns of size $T \times n$ and $\hat{\mathbf{b}} = (b_1, b_2, \dots, b_T)$ be the vector of samples of returns of benchmark (B) of size T . Here T is the number of historical observations which we assume is same for both portfolio and benchmark returns and n is the number of assets from which the portfolio is constructed. Scenario probabilities are fixed to $\frac{1}{T}$ for the whole paper and it is not within its scope to discuss robustness of the probabilities. It could be, however, possible to generalize the model proposed further with any given probabilities of scenarios as well as different history lengths of portfolio and benchmark. The interval of determination of stochastic dominance is cut between the minimum of all observed values:

$$\underline{x} = \min \left(\min_{i,j} \hat{\mathbf{R}}_{i,j}, \min_k \hat{\mathbf{b}}_k \right), \quad i, j, k = 1, \dots, T \quad (3)$$

and the maximum of observed values:

$$\bar{x} = \max \left(\max_{i,j} \hat{\mathbf{R}}_{i,j}, \max_k \hat{\mathbf{b}}_k \right), \quad i, j, k = 1, \dots, T \quad (4)$$

Test hypothesis that the returns of portfolio P stochastically dominates the returns of benchmark B at order s is defined as follows:

$$\begin{aligned} H_0 : D_P^s(x) &\leq D_B^s(x) \quad \forall x \in [\underline{x}, \bar{x}] \\ H_1 : D_P^s(x) &> D_B^s(x) \quad \text{for at least one } x \in [\underline{x}, \bar{x}] \end{aligned} \quad (5)$$

It can be shown that empirical case of function $D_X^s(x)$ has the following form (McFadden (1989)):

$$\hat{D}_X^s(x) = \frac{1}{l} \sum_{i=1}^l \frac{1}{(s-1)!} 1(X_i \leq x)(x - X_i)^{s-1} \quad (6)$$

where l is the number of observations of a variable X . It has to be noted here that this form applies to all $s \in \mathbb{Z}^+$ because by setting $s = 1$ in Eq. (6) we gain Eq. (2).

The stochastic dominance test hypotheses in empirical form is than the following:

$$\begin{aligned} H_0 : \hat{D}_P^s(x) &\leq \hat{D}_B^s(x) \quad \forall x \in [\underline{x}, \bar{x}] \\ H_1 : \hat{D}_P^s(x) &> \hat{D}_B^s(x) \quad \text{for at least one } x \in [\underline{x}, \bar{x}] \end{aligned} \quad (7)$$

The test statistic $\hat{S}^s$ for the order of dominance s of this test using empirical functions is determined in the following way (Linton et al. (2005)):

$$\hat{S}^s = \sqrt{T} \sup_{x \in [\underline{x}, \bar{x}]} [\hat{D}_P^s(x) - \hat{D}_B^s(x)] \quad (8)$$

To determine the critical value of test statistic, the bootstrap method is used in this paper. The test statistic for time series is constructed according to Linton et al. (2005) as follows. We shall take k ($k < T$, $k > 1$) consecutive observations of portfolio and benchmark starting from the same timestamp i , calculate $\hat{S}^s$ for each of the k subsamples, and construct a bootstrapped distribution of the test statistic $\hat{S}^{s*}$. A bottleneck of this method is that the size of bootstrapped observations is equal to $T - k + 1$, which might not be sufficient for small samples. As we wish to obtain the test statistic for our portfolio optimization problem, the following method to obtain the bootstrapped statistic $\hat{S}^{s*}$ is proposed.

Let $\hat{\Theta}_i$ be the vector of returns of portfolio Θ_i with randomly generated weights for $i = 1, \dots, n_b$, where n_b is the number of bootstrapped random portfolio repetitions, $j \in \{1, \dots, T - k + 1\}$ be the starting index of a subsample for a given randomly generated portfolio, $\hat{\Theta}_{i,j} = \{\hat{\Theta}_{i,j}, \hat{\Theta}_{i,j+1}, \dots, \hat{\Theta}_{i,j+k-1}\}$ be the subsample of size k of returns of portfolio Θ_i , and $\hat{\mathbf{b}}_j = \{\hat{b}_j, \hat{b}_{j+1}, \dots, \hat{b}_{j+k-1}\}$ be the subsample of size k of returns of benchmark B . Bootstrapped test statistic for $\hat{\Theta}_{i,j}$ is calculated as follows:

$$\hat{S}_{i,j}^s = \sqrt{k} \sup_{x \in [\underline{x}, \bar{x}]} [\hat{D}_{\hat{\Theta}_{i,j}}^s(x) - \hat{D}_{\hat{\mathbf{b}}_j}^s(x)], \quad i = 1, \dots, n_b, j = 1, \dots, T - k + 1. \quad (9)$$

In this way, $n_b * k$ bootstrapped samples of $\hat{S}^s$ are obtained. The test statistic S_α^s is determined from bootstrap values at significance level α as the $(1 - \alpha)$ -quantile of bootstrapped values of $\hat{S}^s$. $S_{\alpha=100\%}^s = 0$ is defined enforcing the strict stochastic dominance constraint. Formula (9) is used for $\alpha < 1$. The supremum is replaced with the maximum over a finite set of grid points G , as it is not possible to evaluate an infinite number of values on a continuous interval. How to form such a set so that the stochastic dominance properties are not lost is further discussed in the following section.

3. Stochastic dominance in portfolio optimization

This subsection describes the current methodology of how stochastic dominance is implemented into portfolio optimization problems. Stochastic dominance constraint portfolio optimization problem (SD) is defined as follows:

$$\begin{aligned} \max \quad & \mathbb{E}[u(R(\mathbf{w}))] \\ \text{s.t.} \quad & X \succeq_{(s)} B, \\ & \mathbf{w} \in W, \\ & W = \{\mathbf{w} \in \mathbb{R}^n, \sum_{i=1}^n w_i = 1, w_i \geq 0, i = 1, \dots, n\}, \end{aligned} \quad (10)$$

where X is the random return of portfolio determined by weights $\mathbf{w}$, B is the random return of benchmark, n is the number of assets in portfolio and u is any concave utility function of portfolio returns.

Stochastic dominance (SD) has emerged as an essential tool for integrating investor preferences directly into portfolio optimization by evaluating complete return distributions, extending beyond classical mean-variance approaches. First-order stochastic dominance (FSD), originally applied to portfolio contexts by Bawa et al. (1985), ensures universal preference by all non-satiating investors but creates non-convex optimization problems typically solved via mixed-integer programming (Noyan & Ruszczyński, 2008; Post & Kopa, 2009). These computational demands limit the widespread use of FSD in practical applications.

Second-order stochastic dominance (SSD) has received considerable attention due to its alignment with risk-averse investor preferences and computational tractability. Dentcheva and Ruszczyński (2006) significantly advanced SSD portfolio optimization by converting dominance conditions into linear constraints derived from expected shortfall as a follow up of work of Ogryczak and Ruszczyński (1999). Subsequent empirical studies demonstrated the superior performance and natural asset sparsity of SSD-based models compared to classical benchmarks and index-tracking portfolios (Post and Kopa (2013), Roman et al. (2013)).

Higher-order stochastic dominance criteria, such as third-order stochastic dominance (TSD), account for investor preferences related to skewness. Although these approaches have been less frequently studied, recent research highlights their potential to enhance portfolio selection for specific investor classes (Ali et al. (2025), Post and Kopa (2017)).

Robustness extensions have been further explored to mitigate sensitivity to estimation errors and distributional uncertainty. Recent developments include distributionally robust SSD formulations leveraging Wasserstein ambiguity sets (Hosseini-Nodeh et al. (2022), Mei et al. (2021)), interval-based robustness models addressing uncertain asset returns ((Sehgal & Mehra, 2020)), robust optimization models using contamination techniques ((Dupačová & Kopa, 2014)), and empirical likelihood-based stochastic dominance formulations ((Post et al., 2018)).

To address limitations of strict dominance criteria in noisy empirical contexts, Leshno and Levy (2002) proposed the concept of Almost Stochastic Dominance (ASD), allowing for minor violations of SD constraints. Building upon this methodology, this paper introduces a novel approach employing a Kolmogorov–Smirnov (KS)-type test statistic, inspired by the methodology of Linton et al. (2005), as a robust decision criterion. By constraining the maximum violation of stochastic dominance within a statistically insignificant bound, this method offers principled tolerance for sampling noise while preserving the core advantages of SD. Proposed framework thus presents a practical and statistically grounded enhancement to existing SD-based portfolio optimization models.

To the author's knowledge, the application of statistical test metrics directly in stochastic dominance constraints to manage outliers and empirical irregularities remains unexplored. This paper introduces a novel approach employing a Kolmogorov–Smirnov (KS)-type test statistic, building upon the methods of Linton et al. (2005), as a robust criterion within SD constraints. This innovative test-statistic-based approach allows statistically insignificant dominance violations.

3.1. Robust stochastic dominance constraint portfolio optimization problem

In this paper, the objective function is the maximization of the expected return of a portfolio, and a deterministic version of the stochastic dominance test statistic is used. The robust stochastic dominance-against-outliers, general-order- s constraint portfolio optimization model, called in this paper s -RSD, is formulated as follows:

$$\begin{aligned} \max \quad & \mathbb{E}[\hat{\mathbf{R}} \mathbf{w}] \\ \text{s.t.} \quad & \hat{S}^s(\mathbf{w}) \leq S_\alpha^s, \\ & \mathbf{w} \in W. \end{aligned} \quad (11)$$

where $\hat{S}^s(\mathbf{w})$ denotes the test statistic of the portfolio return distribution determined by the vector of asset weights $\mathbf{w}$ according to Eq. (8), and S_α^s denotes the critical value obtained by the bootstrap method according to Eq. (9). By setting $\alpha = 100\%$, we gain a non-robust strict version of stochastic dominance, where we strictly require $\hat{D}_p^s(x) - \hat{D}_B^s(x) \leq 0$ for any given order s . In the next section, we show that, by dropping the disturbance term from the stochastic-dominance constraint, one recovers the classical models found in the literature. This formulation thus acts as a unifying generalization of the core stochastic-dominance portfolio-optimization approaches.

To follow expected utility theory, one must add a condition on means ($\mathbb{E}[\hat{\mathbf{R}}\mathbf{w}] \geq \mathbb{E}[\hat{\mathbf{b}}]$) for higher orders $s \geq 3$ (Levy (1992), Whitmore (1970)); but, as in this paper the expected return $\mathbb{E}[\hat{\mathbf{R}}\mathbf{w}]$ is maximized, one can compare the optimum with the expected return of the benchmark, and if the benchmark's expected return is higher, the portfolio can be rejected under the given stochastic dominance order. Therefore, the mean condition is not accounted for in this paper.

To make the abstract model (11) available for computation, the following section discusses selection of an evaluation grid G for 3 different cases ($s = 1$, $s = 2$ and $s \geq 3$) so that $\sup_{x \in [\underline{x}, \bar{x}]} [\hat{D}_p^s(x) - \hat{D}_B^s(x)] = \max_{x \in G} [\hat{D}_p^s(x) - \hat{D}_B^s(x)]$, and it presents an exact finite-dimensional reformulation of model (11) for these cases.

1-RSD model formulation

By setting $s = 1$ in (6), we get $\hat{D}_p^1(x) = \frac{1}{T} \sum_{i=1}^T 1(\hat{\mathbf{r}}_i \mathbf{w} \leq x) = \hat{F}_p(x)$, where $\hat{\mathbf{r}}_i$ denotes the i -th row of the return matrix $\hat{\mathbf{R}}$, containing returns of assets at time i , and, analogously, $\hat{D}_B^1(x) = \hat{F}_B(x)$. As a reminder, in (8), we enforce $\sup_{x \in [\underline{x}, \bar{x}]} [\hat{F}_p(x) - \hat{F}_B(x)] \leq \frac{S_\alpha^1}{\sqrt{T}}$.

Function $f(x) = \hat{F}_p(x) - \hat{F}_B(x)$ jumps by $\frac{1}{T}$ at each $x = \hat{\mathbf{r}}_i \mathbf{w}$ and by $-\frac{1}{T}$ at each $x = b_i$. A global maximum can therefore occur only just before every downward jump, i.e., at $x = b_i - \epsilon$, where $\epsilon > 0$ is a small number. If there are one or more upward jumps at point i , it does not violate the constraint, because for the sorted values of B , the value difference $\hat{F}_p^1(b_T) - \hat{F}_B^1(b_T) \leq 0 \leq \frac{S_\alpha^1}{\sqrt{T}}$, and all points before $b_T - \epsilon$ are evaluated. To assess this, the Eq. (6) must be reformulated as follows:

$$\hat{D}_p^{1'}(x) = \frac{1}{T} \sum_{i=1}^T 1(\hat{\mathbf{r}}_i \mathbf{w} < x) \quad (12)$$

and likewise for $\hat{D}_B^{1'}(x)$. In this form, the supremum over $[\underline{x}, \bar{x}]$ is equal to maximum:

$$\sup_{x \in [\underline{x}, \bar{x}]} f(x) = \max_{1 \leq i \leq T} [\hat{D}_p^{1'}(b_i) - \hat{D}_B^{1'}(b_i)], \quad (13)$$

and one needs only T benchmark breakpoints to evaluate the maximum.

Let $G_1 = \{b_1, \dots, b_T\}$ be the evaluation grid for the 1-RSD model. Define binaries $z_{ij} \in \{0, 1\}$ for $i, j = 1, \dots, T$ as $z_{ij} = 1 \iff \hat{\mathbf{r}}_i \mathbf{w} < b_j$. Then the 1-RSD MILP model is reformulated as follows:

$$\begin{aligned} \max \quad & \mathbb{E}[\hat{\mathbf{R}}\mathbf{w}] \\ \text{s.t.} \quad & \hat{\mathbf{r}}_i \mathbf{w} \leq b_j - \epsilon + M(1 - z_{ij}), & i, j = 1, \dots, T, \\ & \hat{\mathbf{r}}_i \mathbf{w} \geq b_j - Mz_{ij}, & i, j = 1, \dots, T, \\ & \underbrace{\frac{1}{T} \sum_{i=1}^T z_{ij} - \hat{D}_B^{1'}(b_j)}_{=\hat{D}_p^{1'}(b_j)} \leq \frac{S_\alpha^1}{\sqrt{T}}, & j = 1, \dots, T, \\ & \mathbf{w} \in W, \end{aligned} \quad (14)$$

with $M = \bar{x} - \underline{x} + \epsilon$.

However, the resulting MILP formulation is not exact due to the strictly positive ϵ . An exact MILP formulation, excluding robustness considerations, can alternatively be derived based on the Value-at-Risk (VaR) approach (see Kopa and Kozmík (2025)) and is presented below:

$$\max \quad \mathbb{E}[\hat{\mathbf{R}}\mathbf{w}]$$

$$\text{s.t.} \quad \hat{\mathbf{R}}\mathbf{w} \geq \Lambda B,$$

$$\begin{aligned} \sum_{i=1}^T \lambda_{i,j} &= 1, & j = 1, \dots, T, \\ \sum_{j=1}^T \lambda_{i,j} &= 1, & i = 1, \dots, T, \\ \lambda_{i,j} &\in \{0, 1\}, & i, j = 1, \dots, T, \\ \mathbf{w} &\in W. \end{aligned} \quad (15)$$

This formulation is not, however, suitable for the generalized evaluation function in the Genetic Algorithm, and it is not applicable for implementing the outlier-robustness parameter S_α^1 . One could implement it by letting some of the row and column sums of $\lambda_{i,j}$ be equal to 0, thereby relaxing the strict stochastic dominance constraint.

While the relaxation in model (14) means the dominance condition is no longer exact, the benchmark-based MILP with a small $\epsilon > 0$ remains a close approximation, and supports robust extensions. Therefore, this relaxed formulation will be used in the remainder of the study. Furthermore, the formulation of stochastic dominance in (13) is equivalent to that in the Kopa and Kozmík (2025) model, and it can be used in the Genetic Algorithm defined later.

2-RSD model formulation

By setting $s = 2$ in (6) we obtain $\hat{D}_p^2(x) = \frac{1}{T} \sum_{i=1}^T \max\{x - \hat{\mathbf{r}}_i \mathbf{w}, 0\}$, and analogically for $\hat{D}_B^2(x)$. From (8) we enforce $\sup_{x \in [\underline{x}, \bar{x}]} [\hat{D}_p^2(x) - \hat{D}_B^2(x)] \leq \frac{S_\alpha^2}{\sqrt{T}}$. In this case, the supremum occurs at one of the benchmark points due to the linear behaviour of $\hat{D}_p^2(x)$ and $\hat{D}_B^2(x)$. This has already been proven (see, e.g., (Ali et al., 2025; Post & Kopa, 2017)), and the problem can be formulated as a linear programming (LP) model.

Let $G_2 = \{b_1, \dots, b_T\}$. Define $t_{ij} \geq 0$ for $i, j = 1, \dots, T$ as $t_{ij} \geq b_j - \hat{\mathbf{r}}_i \mathbf{w}$, $t_{ij} \geq 0$. These replace the $\max\{x - \hat{\mathbf{r}}_i \mathbf{w}, 0\}$ in $\hat{D}_p^2(b_j)$. The exact 2-RSD LP model is then:

$$\begin{aligned} \max \quad & \mathbb{E}[\hat{\mathbf{R}}\mathbf{w}] \\ \text{s.t.} \quad & t_{ij} \geq b_j - \hat{\mathbf{r}}_i \mathbf{w}, & i, j = 1, \dots, T, \\ & t_{ij} \geq 0, & i, j = 1, \dots, T, \\ & \frac{1}{T} \sum_{i=1}^T t_{ij} - \hat{D}_B^2(b_j) \leq \frac{S_\alpha^2}{\sqrt{T}}, & j = 1, \dots, T, \\ & \mathbf{w} \in W. \end{aligned} \quad (16)$$

This LP enforces $\max_{z \in G_2} [\hat{D}_p^2(z) - \hat{D}_B^2(z)] \leq S_\alpha^2$, equivalent to the original supremum constraint. By setting $\alpha = 0$ we get the equivalent model derived from expected shortfall by Dentcheva and Ruszczyński (2006).

The specific case of the second-order from the general formulation ends up as a linear program. However, it has to be acknowledged that the literature on portfolio optimization under SSD constraints includes a wide range of more efficient methods. For example, Fábíán et al. (2011) developed cutting-plane algorithms tailored for SSD-constrained models to improve computational efficiency. More recent contributions leverage distributionally robust optimization, such as the Wasserstein-metric-based formulations of Hosseini-Nodeh et al. (2022) and Mei et al. (2022), which can be reformulated as conic programs or solved via decomposition into semi-infinite programs. A broad overview of numerically efficient methods for SSD constraints, including dual methods and event-cutting-plane algorithms, is presented in Dentcheva and Ruszczyński (2024, Chapter 7). A comprehensive review of these methods is beyond the scope of this paper, but they are noted here to provide visibility to the broader literature.

s-RSD model formulation for $s \geq 3$

By setting $s \geq 3$ in (6) we obtain $\hat{D}_p^s(x) = \frac{1}{T} \sum_{i=1}^T \max\{x - \hat{r}_i w, 0\}^s$, and analogically for $\hat{D}_B^s(x)$. From (8) we enforce

$$\sup_{x \in [\underline{x}, \bar{x}]} [\hat{D}_p^s(x) - \hat{D}_B^s(x)] \leq \frac{S_\alpha^s}{\sqrt{T}}. \quad (17)$$

It is not possible to replace the supremum in (8) with the max function over a finite grid due to the polynomial behaviour of $D_p^s(x)$ and $D_B^s(x)$. However, so-called superconvex conditions can enforce the stochastic dominance constraints over the whole continuous interval while tightening the feasible set formed by those constraints. These conditions force the difference between two evaluation points on a grid to lie below the tangent line.

The tangent line is formed by setting a small increment in $\hat{D}_p^s(x)$ via multiplication by $(1 + \epsilon)$, where $\epsilon \geq 0$. (Bawa et al., 1985) fixed the value of ϵ for all evaluation points. This approach was later improved by (Post & Kopa, 2017) by allowing individual ϵ values for different observations. The following formulation is stricter in tightening the set of feasible solutions due to the generalization via parameter S_α^s , and it can be shown that when these conditions hold, then (17) holds. Furthermore, by increasing the number of evaluation points and decreasing the step size between them, the superconvex solution converges to the exact SD solution.

The following definition of superconvex inequalities is formulated for general $s > 0$ and outlier-deviation parameter S_α^s .

Let $b_1 < b_2 < \dots < b_T$ be the ordered benchmark returns, and define a finite grid

$$G_{s \geq 3} = \{g_1 < g_2 < \dots < g_m\} \subset [\underline{x}, \bar{x}] \quad (18)$$

such that

1. $g_1 > b_1$,
2. $g_m = b_T$.

Define constraints tightening factors ϵ_i , for $i = 2, \dots, m$,

$$\epsilon_i = \frac{\hat{D}_B^s(g_i)}{\hat{D}_B^s(g_{i-1}) + \hat{D}_B^{s-1}(g_{i-1})(g_i - g_{i-1})} - 1 \quad i = 2, \dots, m. \quad (19)$$

The superconvex inequalities are

$$\begin{aligned} \hat{D}_p^s(g_1) &\leq \frac{S_\alpha^s}{\sqrt{T}}, \\ (1 + \epsilon_i) \hat{D}_p^s(g_i) &\leq \hat{D}_B^s(g_i) + (1 + \epsilon_i) \frac{S_\alpha^s}{\sqrt{T}} \quad i = 2, \dots, m. \end{aligned} \quad (20)$$

Proposition 1 (Superconvex criterion for s th-order dominance). *If for each evaluation point $g_i \in G_{s \geq 3}$ the superconvex inequalities above hold, then*

$$\hat{D}_p^s(x) \leq \hat{D}_B^s(x) + \frac{S_\alpha^s}{\sqrt{T}} \quad \forall x \in [\underline{x}, \bar{x}].$$

Proof. Set

$$f_s(x) = \hat{D}_p^s(x) - \hat{D}_B^s(x).$$

Let's show $f_s(x) \leq \frac{S_\alpha^s}{\sqrt{T}}$ on each region.

(i) First interval $x \in [\underline{x}, g_1]$.

Since $\hat{D}_p^s$ is nondecreasing and $\hat{D}_p^s(g_1) \leq \frac{S_\alpha^s}{\sqrt{T}}$, for all $x \leq g_1$ we have

$$f_s(x) = \hat{D}_p^s(x) - \hat{D}_B^s(x) \leq \hat{D}_p^s(x) \leq \hat{D}_p^s(g_1) \leq \frac{S_\alpha^s}{\sqrt{T}}.$$

(ii) Inductive intervals $x \in [g_{i-1}, g_i]$, $i = 2, \dots, m$.

Define the supporting tangent to $\hat{D}_B^s$ at g_{i-1} :

$$T_i(x) = \hat{D}_B^s(g_{i-1}) + \hat{D}_B^{s-1}(g_{i-1})(x - g_{i-1}),$$

so $T_i(x) \leq \hat{D}_B^s(x)$ on $[g_{i-1}, g_i]$ from convexity of $D_B^s(x)$. By the definition of ϵ_i ,

$$(1 + \epsilon_i) \hat{D}_p^s(g_i) \leq \hat{D}_B^s(g_i) + (1 + \epsilon_i) \frac{S_\alpha^s}{\sqrt{T}} \iff \hat{D}_p^s(g_i) \leq T_i(g_i) + \frac{S_\alpha^s}{\sqrt{T}}.$$

Together with $\hat{D}_p^s(g_{i-1}) \leq \hat{D}_B^s(g_{i-1}) + \frac{S_\alpha^s}{\sqrt{T}}$, convexity of $\hat{D}_p^s$ gives for any $x = \lambda g_{i-1} + (1 - \lambda)g_i$,

$$\hat{D}_p^s(x) \leq \lambda \hat{D}_p^s(g_{i-1}) + (1 - \lambda) \hat{D}_p^s(g_i) \leq T_i(x) + \frac{S_\alpha^s}{\sqrt{T}}.$$

Since $T_i(x) \leq \hat{D}_B^s(x)$, it follows $f_s(x) \leq \frac{S_\alpha^s}{\sqrt{T}}$ on $[g_{i-1}, g_i]$. With this induction the condition (8) holds on the whole interval $[\underline{x}, \bar{x}]$. $\square$

Let:

- $G_{s \geq 3} = \{g_1 < \dots < g_m\} \subset [\underline{x}, \bar{x}]$ be the chosen evaluation grid.
- $w \in W$ be the portfolio weight vector.
- For each $i = 1, \dots, T$ and scenario $j = 1, \dots, T$:
 - $t_{ij} \geq 0$ be an auxiliary variable enforcing

$$t_{ij} \geq g_j - \hat{r}_i w, \quad t_{ij} \geq 0,$$
 so that $\sum_i t_{ij} = T(s-1)! \hat{D}_p^s(g_j)$.
 - $v_{ij} \geq 0$ be an auxiliary power-cone variable satisfying

$$(v_{ij}, 1, t_{ij}) \in \mathcal{K}_\beta, \quad \beta = \frac{1}{s-1},$$
 which enforces $v_{ij} \geq t_{ij}^{s-1}$.
- $\mathcal{K}_\beta = \{(u, v, t) : u \geq 0, v \geq 0, t \geq 0, t^\beta v^{1-\beta} \leq u\}$ be the standard power cone.
- ϵ_i for $i = 2, \dots, m$ be the superconvex tightening factors from Eq. (20).

From Proposition 1, the following conic program is derived to solve the s -RSD superconvex model.

$$\begin{aligned} \max \quad & \mathbb{E}[\hat{\mathbf{R}} w] \\ \text{s.t.} \quad & t_{ij} \geq g_j - \hat{r}_i^\top w, & i, j = 1, \dots, T, \\ & t_{ij} \geq 0, & i, j = 1, \dots, T, \\ & (v_{ij}, 1, t_{ij}) \in \mathcal{K}_\beta, & i, j = 1, \dots, T, \\ & \beta = \frac{1}{s-1}, \\ & \frac{1}{T(s-1)!} \sum_{j=1}^T v_{1j} \leq \frac{S_\alpha^s}{\sqrt{T}}, \\ & \frac{1 + \epsilon_i}{T(s-1)!} \sum_{j=1}^T v_{ij} \leq \hat{D}_B^s(g_i) + (1 + \epsilon_i) \frac{S_\alpha^s}{\sqrt{T}}, & i = 2, \dots, T, \\ & w \in W \end{aligned} \quad (21)$$

3.2. Unified formulation for all orders of dominance

It is possible to further generalize the evaluation of the integrated CDF function so that it is consistent for all orders, because

$$\hat{D}_X^s(x) = \hat{D}_X^{s'}(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{(s-1)!} 1(X_i < x)(x - X_i)^{s-1}, \quad s \geq 2, \quad (22)$$

With this, we can replace the supremum in (8) and form a discrete generalized stochastic dominance constraint:

$$(1 + \epsilon_i^s) \hat{D}_p^{s'}(g_i^s) \leq \hat{g}_i^{s'} + (1 + \epsilon_i^s) \frac{S_\alpha^s}{\sqrt{T}}, \quad i = 1, \dots, m, \quad (23)$$

where m is the number of evaluation points. Table 1 summarizes the key components of the s -RSD model for $s = 1$, $s = 2$, and $s \geq 3$. In each row, we list the finite evaluation grid G , the tightening factors ϵ_i^s , and the adjusted right-hand side (RHS) $g_i^{s'}$.

Table 1
Components of the s -RSD evaluation for different orders.

Order	Grid G	Tightening factor ϵ_i	RHS $g_i^{s'}$
$s = 1$	$G_1 = \{b_1, \dots, b_{m=T}\}$	$\epsilon_i^1 = 0, \quad i = 1, \dots, T$	$g_i^{s'} = D_B^{s'}(g_i^s) \quad i = 1, \dots, T$
$s = 2$	$G_2 = \{b_1, \dots, b_{m=T}\}$	$\epsilon_i^2 = 0, \quad i = 1, \dots, T$	$g_i^{s'} = D_B^{s'}(g_i^s) \quad i = 1, \dots, T$
$s \geq 3$	$G_{s \geq 3} = \{g_1 < \dots < g_m = \bar{x}\} \subset (\min_i b_i, \bar{x})$	$\epsilon_i^s = 0, \quad \epsilon_i^s = \frac{D_B^s(g_i^s)}{D_B^{s-1}(g_{i-1}^s) + D_B^{s-1}(g_{i+1}^s) - D_B^{s-1}(g_i^s)} - 1, \quad i = 2, \dots, m$	$g_1^{s'} = 0, g_i^{s'} = D_B^{s'}(g_i^s) \quad i = 2, \dots, m$

By expressing all orders $s = 1, 2, \geq 3$ in the same discrete grid form with a grid G_s , a vector of tightening factors ϵ^s , and adjusted RHS values $g^{s'}$ we obtain a single element-wise constraint that applies to any s . This uniform representation enables full vectorization of portfolio evaluations, and allows integration with Genetic Algorithm, allowing both simpler implementation and faster runtimes.

3.3. Cardinality constraints

To assess the real investor needs, cardinality constraints are added into the portfolio optimization model. These constraints serve as a control of a proportion of investment that can be taken in a single asset as well as the total count of assets into which investor can allocate his budget. On the opposite site of importance of such a conditions which typically omit transaction costs, there lies a computational complexity of a models when these are implemented into the model of portfolio optimization. Cardinality constraints are defined as follows:

$$\sum_{i=1}^T y_i \leq C, \\ w_{\min} y_i \leq w_i \leq w_{\max} y_i, \quad i = 1, \dots, T, \\ y_i \in \{0, 1\}, \quad i = 1, \dots, T. \quad (24)$$

where $y_i = 1$ indicates that asset i is included in the portfolio and if $y_i = 0$ then $w_i = 0$, $w_{\min}$ and $w_{\max}$ are the minimum and maximum weights allowed for any selected asset, and C is the upper bound on the total number of assets the investor may hold.

Because variables y_i are binary, these constraints increase the computational complexity of the proposed models while aligning them more closely with real-world portfolio construction requirements. There is a dedicated part in the following section on how to incorporate these conditions into the genetic algorithm as a computationally efficient alternative to exact solvers.

4. Genetic algorithm

As noted several times in the previous section, the proposed s -RSD model, particularly once cardinality constraints are included, becomes extremely computationally demanding as both the number of historical returns and the size of the asset universe increase. This section therefore provides a short introduction to the use of metaheuristics in portfolio optimization and presents the Genetic Algorithm applied to the s -RSD model.

The main reason metaheuristics were developed was to solve complex optimization problems. Metaheuristics are useful in cases where an exact algorithm exists to find the optimal solution but whose time complexity prevents real-time computation. Another benefit is that they might find better solutions to non-convex optimization problems than gradient methods, which tend to get stuck in local optima.

Many other metaheuristic approaches have been proposed to solve more complex portfolio optimization problems, such as cardinality-constrained portfolio optimization problems (CCPO). The pioneering paper on CCPO and metaheuristic approaches to solve it was by Chang et al. (2000). The chosen metaheuristics in that article were Tabu Search (TS), Simulated Annealing (SA), and Genetic Algorithm (GA). GA proved to be the best among these three for solving CCPO. The SA approach was further developed in Crama and Schyns (2003), where turnover constraints and trading constraints were added. As cardinality constraints

are used in this paper as well, the use of metaheuristics is already justified by these works.

Fernández and Gómez (2007) proposed a neural network heuristic to solve CCPO using the Hopfield neural network. This approach was then compared to GA, TS, and SA methods proposed by Chang et al. (2000), and none of these algorithms outperformed the others. Rather than finding an alternative to the algorithms used by Chang et al. (2000), Woodside-Oriakhi et al. (2011) further developed those metaheuristics to solve the cardinality-constrained efficient frontier (CCEF) model and showed that their improvements were superior for optimization of 1318 assets, albeit at the cost of increased computational time.

Dhaini and Mansour (2021) proposed a Squirrel Search Algorithm to solve the CCPO and the unconstrained portfolio optimization problem (UPO). This approach was tested on both the mean-variance model and the Sharpe model. The algorithm proved superior for UPO compared to other metaheuristic approaches, and the CCPO algorithm was comparable to existing methods.

A wide variety of nature-inspired metaheuristics applied to portfolio optimization are demonstrated by Faramarzi et al. (2020) with the Marine Predators Algorithm, Doerner et al. (2004) with the Pareto Ant Colony Algorithm, and Kalayci et al. (2017) with the Artificial Bee Colony Algorithm. These works illustrate the diversity of metaheuristic approaches; however, providing a complete overview is beyond the scope of this paper. The Genetic Algorithm is used here, as it has proven reliable in benchmark comparisons across these studies.

The Genetic Algorithm, one of the most popular metaheuristics in the evolutionary algorithm (EA) family, was originally developed by Holland (1975) and further formalized by Holland (1992). It is a nature-inspired algorithm derived from Darwin's theory of natural selection, simulating the behaviour of a biological population. Variations abound, but all share the same core steps: initialization of a population, followed by iterative parent selection, crossover to create offspring, mutation, and replacement of individuals in the population with the new offspring. Each individual is evaluated and assigned a fitness value, which serves as the criterion for parent selection and population update. Stopping criteria typically include reaching a predetermined number of iterations, observing little improvement in the best solution over several iterations, or both.

Each candidate solution is encoded as a chromosome containing all necessary information. In portfolio optimization, this usually means asset weights. Chang et al. (2000) encoded chromosomes in two parts: one part indicating which assets are included, and the other specifying their weights. Anagnostopoulos and Mamanis (2011a) used a hybrid representation proposed by Streichert et al. (2004), where each asset has a binary inclusion variable in the first part of the chromosome and a weight in the second part. It is also possible to encode the chromosome as a one-dimensional vector of asset weights directly (Kaucic et al., 2019), but convergence issues may arise in CCPO due to the difficulty of enforcing feasibility.

In this paper, the entire population of weight vectors is stored in a single $n \times n_{\text{pop}}$ matrix $\mathbf{W} = [\mathbf{w}^{(1)}, \mathbf{w}^{(2)}, \dots, \mathbf{w}^{(n_{\text{pop}})}] \in \mathbb{R}^{n \times n_{\text{pop}}}$, where each column $\mathbf{w}^{(j)}$ is the weight vector for individual j , n is the number of assets as input for optimization and n_{pop} is the size of the population. Initial population $\mathbf{W}$ is generated as follows:

1. Draw independent samples $X_{i,j} \sim \text{Exp}(1)$, for $i = 1, \dots, n$ and $j = 1, \dots, n_{\text{pop}}$.

2. For each column j , let

$$S_j = \{i : X_{i,j} \text{ is among the top } C \text{ values in column } j\}, \quad |S_j| = C.$$

3. Define

$$w_{i,j} = \begin{cases} \frac{X_{i,j}}{\sum_{\ell \in S_j} X_{\ell,j}}, & i \in S_j, \\ 0, & i \notin S_j. \end{cases}$$

Algorithm 1 Weight adjustment (adjust_weights).

Require: $W \in \mathbb{R}^{n \times n_{\text{pop}}}$, bounds $w_{\min}, w_{\max}$, cardinality C
Ensure: $W_{\text{adj}} \in \mathbb{R}^{n \times n_{\text{pop}}}$ with $w_{\min} \leq w_{ij} \leq w_{\max}$ and $\|w^{(j)}\|_0 \leq C$

- 1: $W_{\text{adj}} \leftarrow W$
- 2: $m \leftarrow \lceil 1/w_{\max} \rceil$
- 3: **for** $j = 1, \dots, n_{\text{pop}}$ **do**
- 4: **if** $\#\{i : W_{ij} > 0\} < m$ **then**
- 5: Randomly pick zeros in column j and set them to values in $[w_{\min}, w_{\max}]$
- 6: **end if**
- 7: Normalize column j : $w_{\cdot,j} \leftarrow w_{\cdot,j} / \sum_i w_{ij}$
- 8: **end for**
- 9: Initialize excess, deficit $\in \mathbb{R}^{n_{\text{pop}}}$ to zero
- 10: **for** $j = 1, \dots, n_{\text{pop}}$ **do**
- 11: $\text{excess}[j] \leftarrow 0$, $\text{deficit}[j] \leftarrow 0$
- 12: **for** $i = 1, \dots, n$ **do**
- 13: **if** $w_{ij} > w_{\max}$ **then**
- 14: $\text{excess}[j] \leftarrow \text{excess}[j] + w_{ij} - w_{\max}$
- 15: $w_{ij} \leftarrow w_{\max}$
- 16: **end if**
- 17: **end for**
- 18: Keep the top C weights in column j , zero out the rest, add their sums to $\text{excess}[j]$
- 19: Redistribute $\text{excess}[j]$ proportionally over those i with $0 < w_{ij} < w_{\max}$
- 20: **for** $i = 1, \dots, n$ **do**
- 21: **if** $0 < w_{ij} < w_{\min}$ **then**
- 22: $\text{deficit}[j] \leftarrow \text{deficit}[j] + w_{\min} - w_{ij}$
- 23: $w_{ij} \leftarrow w_{\min}$
- 24: **end if**
- 25: **end for**
- 26: Redistribute $\text{deficit}[j]$ proportionally over those i with $w_{\min} < w_{ij} < w_{\max}$
- 27: **end for**
- 28: **return** W_{adj}

By construction, each column $w^{(j)}$ has exactly C nonzero entries, sums to one, and satisfies the maximum-assets constraint (without loss of generality if $C = n$). These weights, however, need to be adjusted to match any minimal or maximal weight requirements when an asset is included. Algorithm 1 is proposed to handle these conditions. This algorithm efficiently adjusts the weights compared with incorporating violations of the cardinality constraints into the evaluation function of the Genetic Algorithm.

Algorithm 1 works column-wise on the initial weight matrix W . For each candidate portfolio $w^{(j)}$, it first ensures there are at least $\lceil 1/w_{\max} \rceil$ nonzero positions by randomly setting zero weights to $\text{Exp}(1)$ if needed, and then normalizes $w^{(j)}$ so that $\sum_i w_{ij} = 1$. Next, it finds the weights above $w_{\max}$, caps them at $w_{\max}$, and collects the overage as an excess amount. To enforce the cardinality limit C , it then keeps only the top C weights, assigns any excess to the excess pool, and redistributes this excess proportionally among the remaining entries that are strictly between 0 and $w_{\max}$. A symmetric procedure handles weights below $w_{\min}$: any values under $w_{\min}$ are raised to $w_{\min}$, the shortfall is recorded as

a deficit, and this deficit is in turn spread proportionally over entries in $(w_{\min}, w_{\max})$. The result is a sparse vector in each column of W with exactly C nonzeros, each in $[w_{\min}, w_{\max}]$, summing to 1.

After the population $W \in \mathbb{R}^{n \times n_{\text{pop}}}$ is initialized, the portfolio-return matrix $R = \hat{R}W \in \mathbb{R}^{T \times n_{\text{pop}}}$, is formed and the mean-return vector is computed as $\mu = \frac{1}{T} \mathbf{1}^T R \in \mathbb{R}^{n_{\text{pop}}}$. For an evaluation grid $G_s = \{g_i^s\}_{i=1}^m$, the matrix form of empirical D_p^s is calculated for each weight in column j of W . The violation matrix is given by:

$$V = (1 + \epsilon^s) \odot D_p^s - g^{s'} \mathbf{1}_{n_{\text{pop}}}^T - (1 + \epsilon^s) \frac{S_\alpha^s}{\sqrt{T}} \mathbf{1}_{n_{\text{pop}}}^T \in \mathbb{R}^{m \times n_{\text{pop}}}, \quad (25)$$

and the vector of worst-case violations is then obtained as

$$s_j = \max_{1 \leq i \leq m} V_{i,j}, \quad j = 1, \dots, n_{\text{pop}}. \quad (26)$$

Finally, with the scaling factor $\text{scale} = \frac{T}{0.001^{s-1}}$, and penalty parameter $\lambda > 0$, the fitness of individual j is defined by

$$f_j = \mu_j - \lambda (\text{scale} \max(s_j, 0))^2, \quad j = 1, \dots, n_{\text{pop}}. \quad (27)$$

Algorithm 2 Fitness evaluation (evaluate).

Require: Weight matrix $W \in \mathbb{R}^{n \times n_{\text{pop}}}$, returns $\hat{R} \in \mathbb{R}^{T \times n}$, benchmark $b \in \mathbb{R}^T$,
 1: grid $G_s = (g_1, \dots, g_m)$, tightenings $\epsilon \in \mathbb{R}^m$, critical value S_α^s , order s , penalty λ
Ensure: Fitness vector $f \in \mathbb{R}^{n_{\text{pop}}}$

- 2: **Compute portfolio returns:** $R \leftarrow \hat{R}W$
- 3: **Mean returns:** $\mu \leftarrow \frac{1}{T} \mathbf{1}^T R$
- 4: **Compute** $D_p^s \in \mathbb{R}^{m \times n_{\text{pop}}}$ with

$$[D_p^s]_{i,j} \leftarrow \hat{D}_{w_j}^{ts}(g_m)$$
- 5: **Compute right hand side** $g^{s'} \in \mathbb{R}^m$
- 6: **Form violation matrix** $V \in \mathbb{R}^{m \times n_{\text{pop}}}$ by

$$V \leftarrow (1 + \epsilon) \odot D_p^s - g^{s'} \mathbf{1}^T - (1 + \epsilon) \frac{S_\alpha^s}{\sqrt{T}} \mathbf{1}^T$$
- 7: **Constraint violations:** $s \in \mathbb{R}^{n_{\text{pop}}}$, $s_j \leftarrow \max_i V_{i,j}$
- 8: **Scale factor:** $\text{scale} \leftarrow T/(0.001^{s-1})$
- 9: **Fitness:** $f_j \leftarrow \mu_j - \lambda (\text{scale} \max(s_j, 0))^2$
- 10: **return** f

A scale factor is introduced in this way so a unit violation of the s -RSD constraint by one grid step of size 0.001 represents the same baseline cost regardless of order s . Without this normalization, higher orders would naturally amplify the raw violation measure and require retuning of the penalty coefficient λ for each s .

Positive violations are penalized using the quadratic term $\lambda (\text{scale} \max(s_j, 0))^2$. By growing more slowly than a linear penalty initially, this choice smooths out the fitness landscape and avoids excessive changes in the evaluation function that could be caused by small changes in portfolio weights. Small violations therefore remain relatively inexpensive, allowing gentle exploration near the feasibility boundary, while larger breaches incur progressively higher costs, naturally steering the search back toward feasible solutions. This helps the GA converge toward a suboptimal solution. Pseudocode of fitness evaluation is described in Algorithm 2.

In this way, an individual is penalized whenever its test statistic $S^s(w)$ exceeds the critical value $\hat{S}_\alpha^s$; otherwise, no penalty is applied. Any other constraint can be included similarly in the evaluation function, and penalty parameters can be adjusted for each constraint to determine its relative importance in a given model. A Genetic Algorithm can then work as a sort of goal-programming approach toward a feasible solution if no feasible solution exists in the original model.

After evaluating each generated individual, an iterative process starts. The number of iterations without an improvement in the sub-optimal solution k (early stopping) is used in this paper as a termination

criterion of this process. The following process of reproduction is used accordingly to Anagnostopoulos and Mamanis (2011b) as it proved to be satisfying for portfolio optimization in Kaucic et al. (2019) with introduction of a diversity parameter p_{new} which is setting randomly zero weight into the feasible interval value:

- **Parent selection (binary tournament).** The current population W is randomly paired and, in each pair, the individual with the higher fitness f_j is chosen as a tournament winner. Two pools of $\frac{1}{2}n_{pop}$ parents are then sampled (with replacement) from these winners.
- **Crossover.** Uniform crossover is applied: for each parent pair, each child weight w_i is drawn from the corresponding entry of parent 1 or parent 2 with equal probability.
- **Mutation.** Each child is selected for mutation independently with probability p_{mut} . In a chosen child, each weight w_i is then independently mutated with probability μ by setting $w_i \leftarrow \max\{w_i + \mathcal{N}(0, \sigma_m^2), 0\}$.
- **Injection.** Independently, each child undergoes an injection with probability p_{new} : a single zero entry is replaced by a random value in $[w_{min}, w_{max}]$.
- **Normalization and adjustment.** Children are rescaled to sum to one, then passed through weight adjustment Algorithm 1 to enforce $[w_{min}, w_{max}]$ bounds and cardinality C .
- **Fitness evaluation.** The penalized objective $f_j = \mu_j - \lambda(\text{scale max}(s_j, 0))^2$ is recomputed for all offspring.
- **Survivor selection.** Parents and offspring are merged, sorted by fitness f_j , and the top n_{pop} individuals are retained.
- **Early stopping.** The best fitness f_{best} is tracked; if no improvement occurs for k consecutive generations, the Algorithm terminates.

Finally, the individual with highest f is returned as the sub-optimal solution of the given s -RSD. After this iterative process, the best individual with the highest value of f is chosen as a sub-optimal solution of a given s -RSD. The full pseudocode of GA used in this paper is written in Algorithm 3.

The entire population of weight vectors is stored in the $n \times n_{pop}$ matrix W so that every GA operation: fitness evaluation, crossover, mutation, injection, normalization and weight adjustment can be implemented in Python framework as tensor-matrix computations in *TensorFlow*. Aside from the outer loop over generations, no Python-level loops are used: all heavy computations are expressed as vectorized *TensorFlow* kernels that run efficiently on GPU, providing significant computational efficiency compared to scalar or Python-loop implementations.

Parameter tuning

The mutation and injection parameters p_{mut} , μ , σ_m , and p_{new} were tuned via a grid search over $[0, 1]$. The 2-RSD model chosen had $S_{\alpha=1000\%}^2 = 0$, $w_{min} = 0.02$, $w_{max} = 0.2$, and $C = 15$ with a data history of $T = 250$. It is assumed that these parameters will work across different settings of the s -RSD model. Other model versions will be compared further in the next section. For each combination, the GA ran 30 times and the resulting fitnesses f^* were recorded. Rather than simply maximizing the average fitness, the criterion $\bar{f} - 2 \text{std}(f^*)$ was used to favour settings that deliver both high return and low variability in convergence. The best-performing parameter values found were $p_{mut} = 0.25$, $\mu = 0.75$, $\sigma_m = 1$, and $p_{new} = 0.75$. These settings balance exploration (via mutation and injection) with reliable convergence across runs.

Another separate grid search was performed to determine the optimal patience parameter k and population size n_{pop} . Since only two variables are considered, the results are shown in Table 2. The patience parameter varies among 25, 50, 100, and 200, and the population size among 100, 1 000, 10 000, and 60 000. We collected runtime in seconds, as well as the mean and standard deviation of the optimal evaluation value f^* . It is clear that 1 000 individuals are insufficient for proper GA convergence, due to the lower mean and higher standard deviation. The mean and standard deviation for population sizes of 10 000 and 60 000

Algorithm 3 Genetic algorithm for s -RSD portfolio optimization.

Require: Historical returns $\hat{R} \in \mathbb{R}^{T \times n}$, benchmark $b \in \mathbb{R}^T$, grid $e = (g_1, \dots, g_m)$, tightenings $\epsilon \in \mathbb{R}^m$, critical value S_{α}^s ,
 1: GA parameters: n_{pop} , $iters$, p_{mut} , μ , σ_m , p_{new} , k ,
 2: bounds w_{min}, w_{max} , cardinality C , penalty λ , scale factor.
Ensure: Best weight vector $w_{best} \in \mathbb{R}^n$

▷ 1. Initialization

3: Generate initial population $W \in \mathbb{R}^{n \times n_{pop}}$ (sparse weights)
 4: $f \leftarrow \text{EVALUATE}(W)$
 5: $w_{best} \leftarrow \arg \max_j f_j$, $f_{best} \leftarrow \max_j f_j$
 6: $\text{no_improve} \leftarrow 0$
 7: **for** $\text{gen} = 1$ to $iters$ **do**

▷ 2. Parent selection (binary tournament)

8: Randomly pair the n_{pop} columns of W
 9: In each pair retain the column with higher fitness f_j
 10: Sample (with replacement) two parent pools P_1, P_2 , each of size $\frac{1}{2}n_{pop}$

▷ 3. Crossover

11: **for** $\ell = 1$ to $\frac{1}{2}n_{pop}$ **do**
 12: For each $i = 1, \dots, n$, set $\text{child}_{\ell}[i] \leftarrow \begin{cases} P_1[i], & \text{w.p. } 0.5, \\ P_2[i], & \text{w.p. } 0.5. \end{cases}$
 13: **end for**

▷ 4. Mutation

14: **for** each child **do**
 15: **if** $\text{rand}() < p_{mut}$ **then**
 16: **for** $i = 1, \dots, n$ **do**
 17: **if** $\text{rand}() < \mu$ **then**
 18: $w_i \leftarrow \max\{w_i + \mathcal{N}(0, \sigma_m^2), 0\}$
 19: **end if**
 20: **end for**
 21: **end if**
 22: **end for**

▷ 5. Injection

23: **for** each child **do**
 24: **if** $\text{rand}() < p_{new}$ **then**
 25: Select one zero entry and set it to $\text{Uniform}[w_{min}, w_{max}]$
 26: **end if**
 27: **end for**

▷ 6. Normalize & adjust

28: Normalize each child to sum to 1
 29: Apply adjust_weights to enforce $[w_{min}, w_{max}]$ and cardinality C

▷ 7. Evaluate offspring

30: $f_{child} \leftarrow \text{EVALUATE}(\text{Children})$

▷ 8. Survivor selection

31: Concatenate (W, f) and $(\text{Children}, f_{child})$
 32: Select top n_{pop} columns by fitness to form new W, f

▷ 9. Update best & early stopping

33: **if** $\max_j f_j > f_{best}$ **then**
 34: $f_{best} \leftarrow \max_j f_j$
 35: $w_{best} \leftarrow \text{corresponding column of } W$
 36: $\text{no_improve} \leftarrow 0$
 37: **else**
 38: $\text{no_improve} \leftarrow \text{no_improve} + 1$
 39: **end if**
 40: **if** $\text{no_improve} \geq k$ **then break**
 41: **end if**
 42: **end for**
 43: **return** w_{best}

Table 2
GA performance under different patience parameters k and population sizes n_{pop} .

$k \setminus n_{pop}$	Time (s)				mean(f^*)(10^{-4})				std(f^*) (10^{-4})			
	100	1000	10,000	60,000	100	1000	10,000	60,000	100	1000	10,000	60,000
25	2.320	2.180	5.360	23.680	3.498	4.642	4.726	4.732	0.502	0.170	0.008	0.005
50	3.690	3.920	8.900	38.590	3.696	4.678	4.733	4.737	0.513	0.126	0.004	0.003
100	4.260	6.740	17.540	70.000	3.642	4.693	4.733	4.738	0.504	0.093	0.004	0.003
200	6.160	7.880	25.360	190.980	3.686	4.684	4.735	4.740	0.488	0.129	0.004	0.003

with patience above 50 iterations are not significantly different, so the deciding factor is average runtime. Runtime is lower with both a smaller patience threshold and a smaller population size. The parameters used for the remainder of the paper are thus $k = 50$ and $n_{pop} = 10,000$.

5. Empirical results

In the final section of this paper, the effectiveness and behaviour of the proposed s -RSD model and GA are presented. First, there is a discussion on the effectiveness of the GA compared to exact solvers; the effect of selecting the grid G for higher-order analysis follows; then the behaviour of the s -RSD model under different levels of α is compared; and, at the end of this section, there is a comparison of the out-of-sample performance of the proposed model against established models using S&P 500 index and its components data.

Data for this section are taken from the American market, represented by the S&P 500 index and its components. The time range of the data is January 1, 2015 to December 31, 2024, thus encompassing a wide variety of trends and market behaviours during several major global events over the last ten years. Data were downloaded from Yahoo Finance¹ and transformed into daily returns. Stocks with missing values in the selected period were dropped, leaving a total of 467 components for the empirical part of this paper.

5.1. GA performance

Before evaluating the behaviour of the s -RSD model, a separate section of this paper is dedicated to comparing the proposed s -RSD model solved with exact solvers and with the Genetic Algorithm. This section demonstrates the efficiency of the GA and the benefits of applying it instead of exact solvers. The setup of the comparison is as follows. Five different time frames (t) are analysed (selected linearly across the available data), and four different data-history durations T are used in the optimization (20, 60, 125, and 250). These time frames and histories are then applied to the s -RSD model for orders 1–4 with $S_{\alpha=100\%}^s = 0$ for all s , $w_{\min} = 0.02$, $w_{\max} = 0.2$, and $C = 15$. This setup allows for a comparison of the efficiency of mixed-integer linear programming (MILP) and mixed-integer conic programming (MICP) solvers with the proposed GA. The MILP solver used is HiGHS (see (Huangfu & Hall, 2018)), implemented in the Python SciPy library, and the MICP solver is MOSEK (2025) with its Python API, so that all compared solvers operate within the same code framework. The models for exact solvers were introduced earlier in this paper.

To compare the proposed GA with exact solvers, following statistics are collected:

- $\bar{\zeta}$: average proportion of value of expected return of portfolio found by GA (w_{GA}^* to actual optimum of optimal portfolio w^*).
- σ_{ζ} : standard deviation of proportion of value of expected return of portfolio found by GA (w_{GA}^* to actual optimum of optimal portfolio w^*).
- $\bar{\xi}$: average of scaled stochastic dominance statistic s from (26) with $scale = \frac{h}{0.001^{s-1}}$ of portfolio found by GA

- t_{opt} : duration of processing of exact solver
- t_{GA} : average duration of processing of GA
- $\sigma(t_{GA})$: standard deviation of duration of processing of GA

The GA was run 30 times to observe its behaviour and convergence toward the true optimum. The size of the evaluation grids for orders 3 and 4 equals the history length T , and they are distributed linearly. The infeasible GA solutions have been removed from this analysis (2 out of 600 GA runs). Results are presented in Table 3, which focuses on convergence toward the true optimum, and Table 4, which displays the time efficiency of the GA compared with exact solvers. Missing values in both tables indicate either an out-of-memory error in the exact solver or excessive processing time (> 2 hours).

Table 3 reports the convergence behaviour of the GA against the exact-solver benchmarks for orders $s = 1, \dots, 4$. Each cell shows, for a given time point t and history length T , the average return ratio $\bar{\zeta}$, its standard deviation σ_{ζ} , and the mean scaled dominance statistic $\bar{\xi}$ over 30 independent GA runs (infeasible outcomes were discarded). By definition, the exact MILP/MICP solvers attain $\bar{\zeta} = 100\%$. By contrast, as expected, the GA exhibits slight underperformance and increased variance, most notably for first-order dominance ($s = 1$) in time frame 5, where the GA converged on average to the true optimum at 96.67%, with $\sigma_{\zeta} = 1.42\%$. For all other combinations of s and T , the GA recovers at least 98% of the optimum ($\bar{\zeta} \geq 98\%$), with negligible variance ($\sigma_{\zeta} \leq 1\%$). There are cases in which the GA outperforms the exact solvers. This is due to rounding errors, which arise especially in the MOSEK solver for higher orders.

Table 4 compares solution times for the exact solvers and the GA. While the exact MILP/MICP methods slow down dramatically as the history length T increases, sometimes running out of memory or hitting the one-hour time limit, the GA remains fast and predictable. Even at $T = 250$, the GA completes in only a few seconds on average, with only slight variation across runs. This steady performance means the GA can handle larger datasets and finer grid evaluation points for higher orders without running into the time-and-memory bottlenecks that limit exact solvers.

5.2. Effect of density of evaluation grid

In this section, a brief discussion on the selection of evaluation grid density G for order $s \geq 3$ is presented. As Post and Kopa (2017) stated, the superconvex conditions of stochastic dominance converge to the exact SD constraints as the grid is refined. They highlighted in particular the need to include evaluation points in the potentially sparse tails of the portfolio returns distribution. They proposed the evaluation grid $G_{\text{Post}} = \{b_1, b_2, \dots, b_T\}$ consisting of the sorted benchmark returns, with $e_2 = 0$. This grid is compared to linearly distributed evaluation points in the interval $(b_1, \bar{x}]$ with different numbers of points (100, 250, 500, 1000). The same model setup is used as in the previous GA versus exact-solvers comparison, but only the GA on a single time frame is evaluated here.

Results in Table 5 clearly show that as the size of the evaluation grid G increases, the superconvex conditions with tightening factors ϵ become less restrictive and converge to the actual SD constraints. The statement from Post and Kopa (2017) about the evaluation grid is confirmed, as the expected return found by the GA is significantly lower

¹ <https://finance.yahoo.com/>, accessed December 31, 2025

Table 3
GA performance compared to exact solvers.

t	T	Order 1			Order 2			Order 3			Order 4		
		$\bar{\zeta}$ (%)	σ_{ζ} (%)	$\bar{\xi}$	$\bar{\zeta}$ (%)	σ_{ζ} (%)	$\bar{\xi}$	$\bar{\zeta}$ (%)	σ_{ζ} (%)	$\bar{\xi}$	$\bar{\zeta}$ (%)	σ_{ζ} (%)	$\bar{\xi}$
1	20	100.00	0.00	0.00	100.00	0.00	0.00	99.44	0.23	0.00	99.74	0.38	0.00
	60				99.52	0.17	0.00	98.18	0.20	0.00	98.95	0.02	0.00
	125				99.96	0.00	0.00	99.83	0.16	0.01	99.93	0.06	0.00
	250				98.96	0.27	0.01	99.47	0.30	0.00	99.42	0.32	0.00
2	20	100.00	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00	100.00	0.00	0.00
	60	99.94	0.04	0.00	99.90	0.05	0.00	99.99	0.01	0.00	100.09	0.01	0.00
	125				99.59	0.09	0.00	99.97	0.01	0.05	99.96	0.06	0.02
	250				99.29	0.43	0.00	99.93	0.01	0.01	100.10	0.00	0.00
3	20	99.90	0.06	0.00	99.77	0.00	0.00	100.00	0.02	0.00	100.13	0.01	0.00
	60				99.66	0.06	0.00	99.83	0.16	0.05	99.95	0.04	0.00
	125				99.79	0.07	0.00	99.74	0.07	0.00	100.00	0.00	0.00
	250				99.71	0.16	0.00	99.96	0.01	0.00	99.83	0.06	0.00
4	20	98.89	0.40	0.00	99.90	0.01	0.00	99.88	0.00	0.00	99.87	0.10	0.00
	60				99.84	0.02	0.00	99.92	0.01	0.00	99.87	0.05	0.00
	125				99.85	0.18	0.01	99.98	0.03	0.00	100.09	0.11	0.00
	250				99.62	0.19	0.00	99.75	0.22	0.01			
5	20	96.67	1.42	0.07	99.60	0.16	0.00	99.72	0.07	0.00	99.96	0.02	0.00
	60				99.84	0.06	0.00	99.98	0.01	0.01	99.71	0.01	0.00
	125				98.98	0.19	0.00	98.55	0.95	0.00	98.77	0.93	0.00
	250				99.87	0.03	0.01	99.88	0.04	0.00	99.96	0.05	0.00

Table 4
GA time efficiency compared to exact solvers.

t	T	Order 1			Order 2			Order 3			Order 4		
		t_{opt}	t_{GA}	$\sigma(t_{\text{GA}})$	t_{opt}	t_{GA}	$\sigma(t_{\text{GA}})$	t_{opt}	t_{GA}	$\sigma(t_{\text{GA}})$	t_{opt}	t_{GA}	$\sigma(t_{\text{GA}})$
1	20	1.68	4.43	1.42	0.08	2.04	0.64	24.00	11.56	4.67	33.65	7.21	3.46
	60				13.31	7.02	4.12	48.74	5.54	1.86	85.93	6.93	2.77
	125				48.79	9.63	1.99	190.71	7.51	2.86	38.96	7.14	3.11
	250				474.68	8.02	3.70	670.37	8.90	4.35	537.76	8.50	4.81
2	20	1.15	3.85	0.88	0.08	1.76	0.06	2.40	1.80	0.06	2.16	1.78	0.06
	60	82.51	8.48	1.96	5.26	6.16	1.74	19.33	5.60	1.14	22.72	5.43	0.87
	125				102.93	6.95	3.71	280.95	7.40	2.61	188.26	8.31	3.56
	250				4655.45	12.06	2.91	678.90	9.18	2.50	505.24	10.81	2.09
3	20	7.63	14.95	4.97	0.71	4.82	0.44	4.03	5.91	0.71	3.94	5.52	0.82
	60				5.41	5.67	2.39	11.46	5.36	2.74	89.45	6.62	3.46
	125				11.02	6.65	2.48	900.78	7.16	1.45	497.47	2.04	0.05
	250				643.02	9.44	4.15	782.68	8.55	1.86	1032.47	9.22	2.70
4	20	119.67	22.25	10.07	0.13	1.86	0.12	4.25	1.84	0.06	4.40	3.83	2.22
	60				9.16	6.17	2.21	24.24	5.51	2.18	50.59	3.87	2.17
	125				42.89	6.06	2.78	37.07	5.86	0.98	44.21	6.24	1.20
	250				107.42	10.84	4.16	1048.15	11.14	5.08			
5	20	68.17	18.82	9.82	0.53	6.52	2.86	4.74	4.99	1.81	4.70	5.70	1.17
	60				13.78	8.35	2.69	39.84	7.70	2.10	10.99	5.90	1.49
	125				207.98	5.23	2.48	85.21	8.05	3.16	426.75	7.65	4.03
	250				970.01	9.44	2.95	610.06	9.67	4.09	243.84	11.26	3.21

Table 5
Expected portfolio returns (%) for different orders and grid.

s	Post	100	250	500	1000
3	0.2185	0.2862	0.2922	0.2929	0.2931
4		0.2864	0.2924	0.2932	0.2934
5		0.2870	0.2925	0.2931	0.2935

when the grid is sparse, due mainly to the sparse tail in the lowest-return region of the distribution. Moreover, because the GA uses vectorized fitness evaluations rather than explicit reformulations for each grid point, it can handle very large, dense grids without the combinatorial explosion in model size that causes exact solvers to run out of memory.

5.3. Overview of s -RSD behaviour with different levels of significance α

Before moving to the comparison of different models on several datasets, we examine the behavior of the s -RSD model across different significance levels α . We compare the empirical functions $\hat{D}_{\chi}^s(x)$ from Eq. (6) for various levels of α and orders of stochastic dominance. Stocks from the S&P 500 index were chosen for this analysis, with the index itself serving as the benchmark portfolio. It is important to note that the S&P 500 index is structured to mirror market growth. Only stochastic dominance constraints are considered here, without cardinality constraints. A one-year history length ($T = 250$ trading days) is used in the optimization.

Results of this analysis are visualized in Fig. 1. The empirical cumulative distribution functions (CDFs) for the returns of the optimally selected portfolio by the first-order s -RSD model, compared with the empirical CDF of the S&P 500 index returns, are shown in the first row of

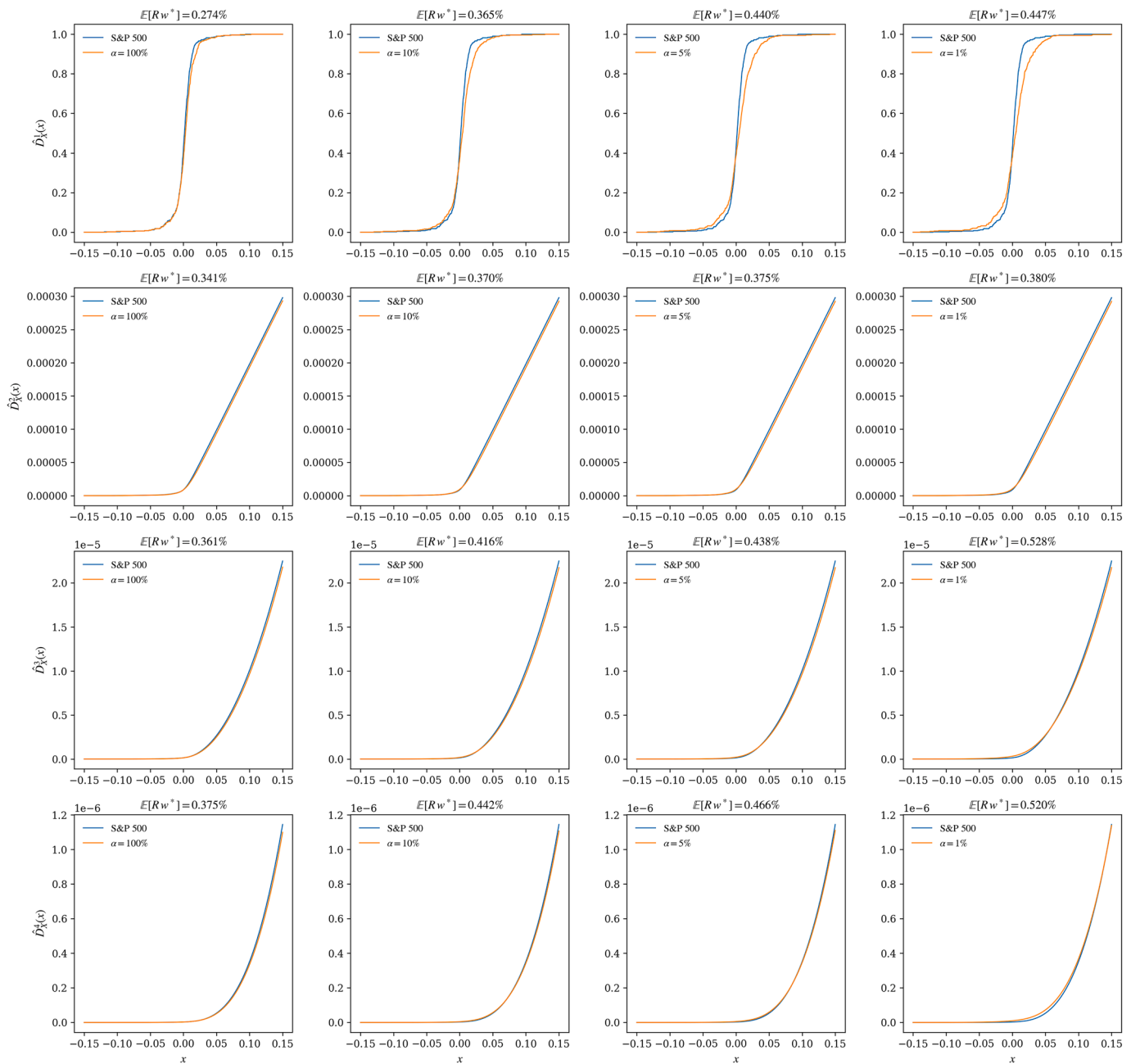


Fig. 1. Empirical functions $\hat{D}_X^s(x)$ of different orders s of S&P 500 daily returns and returns of portfolio optimized with different α levels.

Fig. 1. In the subsequent rows, integrated CDFs for portfolios optimized by 2-RSD, 3-RSD, and 4-RSD are displayed. As anticipated, the differences among these empirical functions increase as the significance level α decreases, which in turn raises S_α^s in the stochastic dominance test. The optimized portfolios for all orders with a fixed $S_{\alpha=100\%}^s = 0$ closely align with the S&P 500 index, as their $\hat{D}_X^s(x)$ functions overlap. This outcome confirms the strongest requirement of stochastic dominance and reflects the well-structured nature of the S&P 500 index in this context, since there is little variation in the portfolios optimized for maximum expected return at the same level of risk as the index.

Incorporating robustness through the significance level α provides confidence intervals for the empirical functions $\hat{D}_X^s(x)$, accommodating minor deviations in the stochastic dominance constraint. This effect is most evident in first-order SD: reducing α causes the empirical CDF of the 1-RSD portfolio to slightly exceed that of the S&P 500 index in the negative-return region. In the range of higher returns, the empirical CDF

of the 1-RSD portfolio is notably lower, indicating potential for higher expected returns with a marginal increase in risk. This pattern holds for higher orders as well. The results demonstrate that allowing a small deviation in risk yields significantly higher expected returns for the constructed portfolios.

5.4. Out of sample performance of different orders of stochastic dominance

As noted at the beginning of this section, the final part of the empirical analysis focuses on the out-of-sample performance of different orders of the s -RSD model and various α levels. To capture each model's behaviour, three testing scenarios are identified: bearish, when the one-year out-of-sample S&P return is below 0%; neutral, when it lies between 0% and 10%; and bullish, when it exceeds 10%. Within each regime, 30 separate points in time are selected, with the intervals between them maximized to capture varied market conditions. A one-year

Table 6
Out-of-sample performance of *s*-RSD model by regime and different levels of α (in %).

		bearish					neutral					bullish				
		S&P	$\alpha = 100$	$\alpha = 10$	$\alpha = 5$	$\alpha = 1$	S&P	$\alpha = 100$	$\alpha = 10$	$\alpha = 5$	$\alpha = 1$	S&P	$\alpha = 100$	$\alpha = 10$	$\alpha = 5$	$\alpha = 1$
Order 1																
Cumulative	μ	-7.55	-1.80	-2.56	-1.60	-0.74	5.70	17.66	21.48	25.04	27.28	19.05	38.83	66.90	73.79	77.52
	σ	6.23	10.02	13.88	15.47	16.23	3.19	16.02	27.30	32.74	37.09	9.61	20.65	50.15	58.37	59.67
	min	-18.75	-21.00	-27.40	-27.20	-34.19	0.75	-10.61	-16.36	-19.87	-23.71	10.24	-2.13	-16.56	-15.35	-15.51
	max	-0.09	17.31	42.26	51.04	51.18	9.96	65.78	107.90	130.04	153.26	53.83	83.40	188.97	237.62	242.87
	$VaR_{0.05}$	18.63	16.22	24.65	24.19	26.35	-1.01	4.77	8.82	9.47	17.22	-10.94	-11.62	-7.98	-8.67	0.62
	$cVaR_{0.05}$	18.69	19.63	26.96	26.37	32.33	-0.83	8.87	12.79	14.78	22.37	-10.58	-4.28	6.02	5.48	11.04
	SR	-153.43	-37.90	-32.83	-23.25	-16.87	116.03	97.76	71.37	70.39	68.17	177.39	178.35	129.41	122.99	126.55
	STARR	-51.10	-19.34	-16.91	-13.64	-8.47	∞	176.53	152.30	155.92	113.03	∞	∞	1077.42	1310.41	684.19
Daily returns	μ	-0.02	0.00	0.01	0.01	0.02	0.03	0.07	0.09	0.10	0.11	0.07	0.14	0.21	0.22	0.24
	σ	1.36	1.64	2.07	2.10	2.20	1.30	1.58	1.94	2.03	2.14	1.11	1.46	2.06	2.16	2.32
	min	-11.98	-13.49	-11.80	-12.61	-13.15	-11.98	-13.10	-12.38	-13.41	-13.47	-11.98	-12.94	-13.62	-14.16	-16.19
	max	9.38	11.02	12.25	11.55	11.49	9.38	16.77	17.71	17.77	19.02	9.38	16.36	18.47	18.47	19.67
	$VaR_{0.05}$	2.33	2.57	3.51	3.52	3.69	1.97	2.30	2.96	3.06	3.23	1.54	2.08	2.98	3.14	3.33
	$cVaR_{0.05}$	3.32	3.91	4.92	5.00	5.20	3.19	3.61	4.49	4.70	4.98	2.68	3.25	4.52	4.75	5.12
	SR	-2.27	-0.23	-0.03	0.16	0.38	1.73	4.19	4.12	4.45	4.54	6.00	8.88	9.79	10.00	9.81
	STARR	-0.93	-0.10	-0.01	0.07	0.16	0.71	1.83	1.78	1.92	1.95	2.49	3.99	4.46	4.55	4.44
Order 2																
Cumulative	μ	-7.55	0.16	-0.45	-0.31	-0.38	5.70	26.03	21.87	21.79	21.77	19.05	45.99	48.08	48.76	51.57
	σ	6.23	12.99	13.92	14.02	14.86	3.19	20.12	20.31	20.57	21.23	9.61	24.22	26.87	27.63	30.08
	min	-18.75	-20.14	-23.48	-24.32	-23.78	0.75	-4.08	-7.71	-8.57	-10.32	10.24	-5.97	-3.96	-6.29	-7.99
	max	-0.09	34.60	38.79	40.25	41.64	9.96	72.08	76.42	78.50	81.55	53.83	88.71	102.37	103.87	107.54
	$VaR_{0.05}$	18.63	17.66	20.66	20.08	20.98	-1.01	0.69	3.59	5.30	5.88	-10.94	-12.13	-9.94	-10.48	-9.87
	$cVaR_{0.05}$	18.69	19.19	22.83	23.07	23.47	-0.83	4.02	5.92	7.24	8.59	-10.58	-2.78	-1.76	-0.85	0.67
	SR	-153.43	-14.20	-17.61	-16.46	-16.02	116.03	119.45	97.85	96.21	93.12	177.39	181.65	171.50	169.25	164.80
	STARR	-51.10	-9.61	-10.74	-10.00	-10.15	∞	597.81	335.62	273.45	230.04	∞	∞	∞	∞	7395.95
Daily returns	μ	-0.02	0.01	0.01	0.01	0.01	0.03	0.10	0.09	0.09	0.09	0.07	0.16	0.17	0.17	0.18
	σ	1.36	1.71	1.81	1.83	1.86	1.30	1.71	1.76	1.78	1.80	1.11	1.61	1.72	1.75	1.82
	min	-11.98	-14.01	-11.64	-11.87	-11.46	-11.98	-14.31	-14.28	-14.08	-14.08	-11.98	-13.71	-14.75	-14.70	-14.39
	max	9.38	11.68	12.67	12.45	12.39	9.38	16.33	16.88	16.79	17.11	9.38	16.63	16.76	17.37	17.59
	$VaR_{0.05}$	2.33	2.72	2.92	2.94	3.02	1.97	2.49	2.57	2.60	2.65	1.54	2.28	2.46	2.52	2.61
	$cVaR_{0.05}$	3.32	4.08	4.30	4.34	4.41	3.19	3.90	4.06	4.10	4.16	2.68	3.58	3.79	3.86	4.01
	SR	-2.27	0.25	0.17	0.21	0.21	1.73	5.53	4.63	4.58	4.52	6.00	9.37	9.17	9.13	9.19
	STARR	-0.93	0.10	0.07	0.09	0.09	0.71	2.42	2.01	1.99	1.96	2.49	4.22	4.15	4.13	4.17
Order 3																
Cumulative	μ	-7.55	1.46	0.96	1.00	1.49	5.70	27.82	25.25	25.67	26.98	19.05	47.39	51.90	55.22	63.38
	σ	6.23	15.28	16.93	17.55	18.42	3.19	22.89	23.01	24.11	25.65	9.61	24.71	27.75	30.11	38.08
	min	-18.75	-18.83	-23.11	-24.15	-25.58	0.75	-9.26	-13.09	-14.31	-17.70	10.24	-3.49	2.11	0.67	-2.28
	max	-0.09	38.02	47.00	49.64	54.01	9.96	77.74	85.16	87.16	90.88	53.83	96.42	105.43	108.60	147.52
	$VaR_{0.05}$	18.63	17.00	19.33	19.61	20.29	-1.01	3.28	2.35	3.22	4.57	-10.94	-11.09	-10.33	-10.15	-11.29
	$cVaR_{0.05}$	18.69	18.02	22.31	23.03	24.85	-0.83	7.10	8.05	8.84	11.63	-10.58	-2.97	-4.92	-4.19	-3.46
	SR	-153.43	-3.55	-6.15	-5.71	-2.79	116.03	112.79	101.06	98.17	97.41	177.39	183.69	179.83	176.75	161.19
	STARR	-51.10	-3.01	-4.66	-4.35	-2.07	∞	363.70	288.99	267.73	214.84	∞	∞	∞	∞	∞
Daily returns	μ	-0.02	0.02	0.02	0.02	0.02	0.03	0.11	0.10	0.10	0.11	0.07	0.16	0.18	0.19	0.21
	σ	1.36	1.76	1.88	1.92	1.97	1.30	1.79	1.86	1.90	1.95	1.11	1.67	1.82	1.91	2.10
	min	-11.98	-14.32	-11.91	-12.07	-12.46	-11.98	-14.20	-14.51	-14.68	-14.53	-11.98	-13.86	-14.15	-14.13	-14.30
	max	9.38	11.44	12.03	12.16	12.42	9.38	16.82	16.89	17.38	17.29	9.38	17.24	15.83	15.97	16.51
	$VaR_{0.05}$	2.33	2.84	3.10	3.19	3.27	1.97	2.62	2.74	2.78	2.89	1.54	2.37	2.63	2.74	2.98
	$cVaR_{0.05}$	3.32	4.20	4.49	4.58	4.70	3.19	4.14	4.32	4.39	4.52	2.68	3.69	3.99	4.18	4.61
	SR	-2.27	0.53	0.46	0.48	0.60	1.73	5.59	5.00	4.99	5.07	6.00	9.33	9.30	9.37	9.54
	STARR	-0.93	0.22	0.19	0.20	0.25	0.71	2.42	2.16	2.15	2.19	2.49	4.21	4.25	4.27	4.34
Order 4																
Cumulative	μ	-7.55	1.85	1.62	1.84	2.19	5.70	28.66	27.25	27.31	28.67	19.05	49.62	54.49	57.83	65.16
	σ	6.23	16.71	18.28	18.65	19.29	3.19	24.64	25.49	25.45	27.13	9.61	25.07	29.01	31.84	39.72
	min	-18.75	-20.04	-23.61	-24.24	-25.51	0.75	-16.50	-20.96	-20.85	-20.71	10.24	-3.48	3.59	2.21	-0.65
	max	-0.09	44.28	50.50	52.72	55.68	9.96	82.46	91.50	94.24	100.64	53.83	98.46	106.68	114.73	157.76
	$VaR_{0.05}$	18.63	17.34	18.17	18.55	18.65	-1.01	3.93	2.65	2.74	4.91	-10.94	-10.55	-10.05	-9.43	-10.09
	$cVaR_{0.05}$	18.69	19.09	22.07	22.80	24.00	-0.83	10.23	11.89	11.88	13.79	-10.58	-2.36	-5.51	-4.75	-3.30
	SR	-153.43	-0.91	-2.06	-0.86	0.98	116.03	108.22	99.06	99.45	98.29	177.39	189.94	180.95	175.34	159.01
	STARR	-51.10	-0.79	-1.71	-0.70	0.79	∞	260.54	212.47	213.00	193.37	∞	∞	∞	∞	∞
Daily returns	μ	-0.02	0.02	0.02	0.02	0.02	0.03	0.11	0.11	0.11	0.11	0.07	0.17	0.18	0.19	0.21
	σ	1.36	1.80	1.90	1.93	1.98	1.30	1.84	1.91	1.94	1.99	1.11	1.70	1.87	1.96	2.13
	min	-11.98	-14.10	-12.20	-12.37	-12.65	-11.98	-13.79	-14.07	-14.11	-13.98	-11.98	-13.58	-13.68	-13.64	-13.76
	max	9.38	11.30	12.00	12.33	12.83	9.38	16.08	17.61	16.89	17.26	9.38	17.12	15.64	15.88	16.19
	$VaR_{0.05}$	2.33	2.91	3.10	3.19	3.28	1.97	2.69	2.81	2.84	2.96	1.54	2.43	2.70	2.82	3.07
	$cVaR_{0.05}$	3.32	4.29	4.53	4.60	4.71	3.19	4.27	4.42	4.49	4.59	2.68	3.75	4.08	4.30	4.69
	SR	-2.27	0.61	0.58	0.64	0.72	1.73	5.57	5.19	5.15	5.24	6.00	9.52	9.44	9.46	9.58
	STARR	-0.93	0.25	0.24	0.27	0.30	0.71	2.41	2.24	2.22	2.27	2.49	4.32	4.33	4.32	4.36

history (250 trading days) of daily returns is used in the optimization, and model performances are evaluated on the following year's out-of-sample data. Cardinality constraints are imposed to reflect realistic conditions: $w_{\min} = 0.02$, $w_{\max} = 0.5$, and $C = 20$.

In comparison with the proposed generalization in the s -RSD model, Kopa and Kozmík (2025) is equivalent to first order with $\alpha = 100\%$ and Dentcheva and Ruszczyński (2006) is equivalent to 2-RSD with $\alpha = 100\%$. The strict version of $\alpha = 100\%$ in Post and Kopa (2017) is represented on the same evaluation grid. As shown earlier, a finer grid yields more precise results. To compare out-of-sample results, the following statistics from Sehgal and Mehra (2020) were collected: (i) μ , the sample mean return; (ii) σ , the sample standard deviation; (iii) min, the minimum return; (iv) max, the maximum return; (v) $\text{VaR}_{0.05}$, the 5% value at risk, defined as the 5% quantile of the return distribution, $\text{VaR}_{0.05} = -\inf\{x \in \mathbb{R} : F_R(x) \geq 0.05\}$; (vi) $\text{CVaR}_{0.05}$, the 5% conditional VaR, defined as the expected return below this threshold, $\text{CVaR}_{0.05} = -\mathbb{E}[R | R \leq \text{VaR}_{0.05}]$; (vii) $\text{SR} = (\mu - r_f)/\sigma$, the Sharpe ratio; and (viii) $\text{STARR} = (\mu - r_f)/\text{CVaR}_{0.05}$ (defined as ∞ if $\text{CVaR}_{0.05} \leq 0$).

The Euribor yearly rate is taken as the risk-free return $r_f = 2\%$ for this paper. The evaluation grid for orders 3 and 4 has size 500 and is linearly distributed, as discussed earlier. Out-of-sample results are reported both as cumulative (final return over one year) and as daily returns across all runs for each model. The S&P 500 index statistics are included for reference. Results are shown in Table 6.

Bearish Regime: Core statistics for the S&P 500 in this regime include an annual average return of -7.55% , annual volatility of 6.23% , a daily mean return of -0.02% , and daily volatility of 1.36% . In the 1-RSD model, at $\alpha = 100\%$, the annual return improves to -1.80% with volatility of 10.02% , cutting losses relative to the index but doubling risk. When α is tightened to 1% , the annual loss falls further to -0.74% and the maximum one-year gain reaches 51.18% , though daily volatility rises to 2.20% , annual volatility to 16.23% , and the worst drawdown hits -34.19% . Yet the Sortino ratio climbs from 0.23 to 0.45 , showing that the extra risk is compensated by positive skew. Under 2-RSD, at $\alpha = 100\%$, the model turns slightly positive with an annual return of $+0.16\%$ and volatility of 12.99% , capping the worst one-year loss at -20.14% and yielding an annual $\text{VaR}_{0.05}$ of 17.66% , which is lower than the index. Lowering α worsens both mean and risk, demonstrating that second-order protection holds only in the strict case. The 3-RSD model delivers positive average returns both annually and daily, with VaR and minima comparable to the index at $\alpha = 100\%$, and relaxing the dominance constraints boosts the return-to-risk ratio and observed maxima. Finally, the 4-RSD model mirrors the third order here, and although more simulations are needed to confirm higher returns, at $\alpha = 1\%$ it delivers the best Sharpe and STARR ratios for both annual and daily returns.

Neutral Regime: Core statistics for the S&P 500 here include an annual average return of $+5.70\%$, annual volatility of 3.19% , a daily mean of 0.03% , and daily volatility of 1.30% . Under strict 1-RSD, the model yields an annual return of 17.66% at volatility of 16.02% and $\text{VaR}_{0.05} = 4.77\%$. At $\alpha = 1\%$, the mean return climbs to 27.28% with a maximum of 153.26% , though $\text{CVaR}_{0.05}$ rises to 22.37% and daily volatility almost doubles to 2.14% . This produces higher average returns as α decreases, but mixed Sharpe and STARR signals—falling for annual returns and rising for daily returns. The 2-RSD model shows that lowering α increases risk and lowers returns, yet its strict form balances return and risk better than the index, offering an attractive alternative in stable markets. Third and fourth orders follow the same pattern: annual returns dip at $\alpha = 10\%$, then recover as α falls, matching the second order with slightly higher returns and risk, and Sharpe ratios that remain similar across orders, especially under strict dominance.

Bullish Regime: Core statistics for the S&P 500 in this regime include an annual average return of $+19.05\%$, annual volatility of 9.61% , a daily mean of 0.07% , and daily volatility of 1.11% . In this regime, 1-RSD performs best: as α decreases, expected returns surge while VaR and CVaR remain low, yielding top annual return-to-risk ratios under strict dominance, though daily Sharpe trends reverse, and extreme upside re-

turns may appeal to bullish investors. Orders 2 through 4 all produce higher expected returns with VaR comparable to the index; although losses beyond VaR grow, they stay modest relative to average returns exceeding 50% . Lowering α consistently boosts returns without destabilizing VaR, suggesting that relaxed stochastic-dominance constraints benefit portfolios in rising markets.

Across all market regimes, higher-order stochastic dominance (s -RSD) models consistently offer enhanced flexibility and improved return profiles compared to the plain S&P 500 benchmark. By tuning the disturbance parameter α , investors can shift the trade-off between downside protection and upside potential: strict versions ($\alpha = 100\%$) of 2nd-4th orders frequently outperform the index in bearish and neutral markets, delivering positive or less negative annual returns with lower VaR and CVaR. In bullish markets, 1-RSD shines, generating sharply increased expected gains and favorable Sharpe ratios, while 2nd-4th orders also capture outsized upside without materially worsening tail risk. Moreover, 3- and 4-RSD models can improve return-to-risk ratios (Sortino/STARR) by exploiting skew and higher-moment preferences.

However, these benefits come with notable bottlenecks. As α decreases, all s -RSD orders exhibit substantially higher annual and daily volatility, and, in extreme cases, larger drawdowns, leveraging some of the gain in average return. The robust stochastic dominance constraint enhances portfolio selection possibilities by permitting controlled deviations from exact dominance, reducing sensitivity to outliers, especially in the low-return zones.

Overall, the results suggest that higher orders of stochastic dominance may present an attractive opportunity, as these models demonstrate robust out-of-sample performance across all market regimes.

6. Conclusion

This paper introduces a novel approach to portfolio optimization by integrating the concept of stochastic dominance constraints. The proposed s -RSD model, coupled with a robust statistical test for stochastic dominance at a given significance level α , presents an efficient strategy for balancing the maximization of returns with the minimization of investment risk. It generalizes traditional strict stochastic dominance frameworks by allowing controlled, statistically insignificant violations, thereby enhancing flexibility and practical robustness in portfolio construction.

The application of a Genetic Algorithm (GA) as a solver for this non-linear optimization problem further enhances the model's practicality. The GA's adaptability in handling various constraints and its ability to escape local optima make it an ideal choice for solving complex portfolio optimization problems. This paper's methodology underscores the efficiency of combining robust statistical tests with evolutionary algorithms in financial decision-making. Furthermore, because the s -RSD model is a generalization, this GA can be applied to established models in the literature. It also allows for the integration of cardinality constraints with minimum and maximum investment limits per asset. An empirical study proved that this algorithm converges efficiently in a fraction of the time required by exact solvers, enabling the inclusion of additional historical observations in the model.

Empirical results across market regimes demonstrate the flexibility and effectiveness of the s -RSD models. In bearish market conditions, where the S&P 500 posts an annual return of -7.6% , strict 1-RSD cuts losses to -1.8% (and to -0.7% at $\alpha = 1\%$), while 2-RSD delivers a positive return ($+0.2\%$) with lower VaR than the index. In the neutral regime, where the index returns 5.7% , strict 1-RSD boosts returns to 17.7% (rising to 27.3% at $\alpha = 1\%$), and 2nd-4th orders offer gains with balanced risk. In the bullish regime, where the S&P 500 returns 19.1% , 1-RSD at low α drives annual gains above 50% without materially increasing tail risk, and higher orders capture upside. Across all regimes, lowering α shifts the trade-off toward greater upside at the cost of higher volatility, while strict higher-order constraints consistently improve return-to-risk ratios (Sharpe, Sortino, STARR) relative to the benchmark.

The flexibility of the s -RSD model is particularly evident in its ability to adapt to varying levels of investor risk tolerance, as indicated by the significance level α . This feature allows for tailored portfolio strategies that align with individual investment goals, whether prioritizing the minimization of risk or the maximization of returns. The ability to choose an order of stochastic dominance simply by setting the parameter s is a major advantage of this method compared to models focused on a single order.

Despite its strengths, this study relies on several simplifying assumptions. Past returns are treated as stationary with equal scenario weights, a finite bootstrap sample, and a discretized grid used to approximate a continuous dominance constraint. Both of which may misestimate true risk. The Genetic Algorithm, while fast, offers no global-optimality or turnover guarantees. Finally, empirical tests are confined to U.S. equities over a one-year horizon; extending to multi-asset portfolios, dynamic probability schemes, and longer or rolling windows would be needed to fully validate the approach.

Building upon the s -RSD framework developed in this paper, future work could explore robust approaches such as adding constraints to guard against errors in return estimates, using sets of possible return distributions to enhance stability, or integrating worst-case distribution robustness. There is an interesting pattern for investors in higher orders of stochastic dominance in the empirical analysis of this paper. It might be interesting to conduct a thorough empirical analysis focusing on data from different markets and observe behaviour even for higher orders. Integrating machine learning tools, like deep neural networks or gradient-boosted trees, within the Genetic Algorithm could help the evolutionary process find strong portfolios more quickly.

In conclusion, the integration of robust stochastic dominance constraints in portfolio optimization, as demonstrated in this paper, offers a flexible approach to investment decision-making. By effectively combining statistical robustness with the computational power of Genetic Algorithms, this methodology provides way to achieve more precise and adaptive portfolio management strategies relevant for both individual investors and financial institutions. As financial markets continue to evolve, such approaches will be helpful in navigating the complexities of investment management.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT (OpenAI) in order to improve the clarity, grammar, and academic style of the manuscript text. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Jakub Neugebauer: Writing – review & editing, Writing – original draft, Project administration, Investigation, Conceptualization.

Declaration of competing interest

I confirm that I am the sole author of this work, that it represents original research not under consideration elsewhere, and that I have no conflicts of interest. During the preparation of this work the author used ChatGPT in order to translate and correct the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. Sincerely, Jakub Neugebauer

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